

Project Design Phase-II

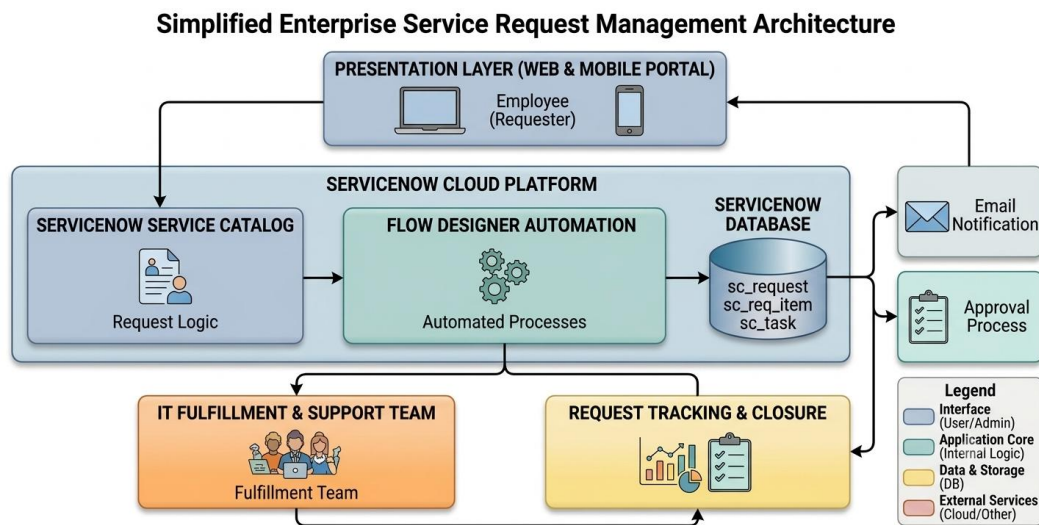
Technology Stack (Architecture & Stack)

Date	6 September 2026
Team ID	Individual Project
Student Name	Samudrala Sujana
Project Name	An Employee Requests Installation of Licensed Software through the Service Catalog
Maximum Marks	4 Marks

Technical Architecture:

The proposed solution uses ServiceNow as a cloud-based platform to automate the complete lifecycle of licensed software installation requests. Employees interact with a Service Catalog item, and the request is processed using ServiceNow Flow Designer, approval actions, catalog tasks, email notifications, and ServiceNow platform tables.

Architecture Flow Diagram



Architecture Description:

- **User Interface:** Employees access the ServiceNow Service Portal and submit a licensed software installation request.
- **Application Logic:** Flow Designer orchestrates the request, approval process, catalog task creation, and email notification.
- **Data Storage:** ServiceNow stores request data in the Service Catalog tables such as **sc_request**, **sc_req_item**, and **sc_task**.
- **External Interfaces:** Email delivery is handled through ServiceNow's configured email capability.
- **Infrastructure:** The solution runs on the ServiceNow cloud platform; no separate local application server or database is required.

Table-1: Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Employees submit and track licensed software installation requests through the Service Portal and Service Catalog.	ServiceNow Service Portal / Service Catalog
2	Application Logic-1	Automates the request lifecycle when a catalog item is requested.	ServiceNow Flow Designer
3	Application Logic-2	Routes the request through the required approval process.	Ask for Approval – Flow Designer Action
4	Application Logic-3	Creates a fulfillment task and assigns it to the Software Support Team.	Create Catalog Task – Flow Designer Action
5	Database	Stores requests, requested items, catalog tasks, variables, and workflow-related records.	ServiceNow Platform Database
6	Cloud Database	Managed cloud storage used internally by the ServiceNow instance.	ServiceNow Cloud Platform
7	File Storage	Optional storage for attachments related to the software request.	ServiceNow Attachments
8	External API-1	No external API is required for the current implementation.	Not Applicable
9	External API-2	No external API is required for the current implementation.	Not Applicable
10	Machine Learning Model	Machine learning is not required for the current workflow automation project.	Not Applicable
11	Infrastructure (Server / Cloud)	The application is deployed and executed as a cloud-based ServiceNow solution.	ServiceNow Cloud Infrastructure

Technology Stack Summary

Layer	Technology / Component
Platform	ServiceNow
Request Interface	Service Catalog / Service Portal
Workflow Automation	Flow Designer
Approval Management	Ask for Approval Action
Task Management	Catalog Task (sc_task)
Request Tables	sc_request and sc_req_item
Notifications	ServiceNow Send Email Action
Hosting	ServiceNow Cloud

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology / Implementation
1	Open-Source Frameworks	The project is primarily configured using ServiceNow's low-code platform and does not depend on a separate open-source application framework.	ServiceNow Low-Code / No-Code Configuration
2	Security Implementations	Role-based access control and ServiceNow authentication restrict access to requests, approvals, and fulfillment tasks.	ServiceNow ACLs, Roles and Authentication
3	Scalable Architecture	The cloud-based workflow can support increasing numbers of employees and software installation requests without redesigning a local server architecture.	ServiceNow Cloud Platform
4	Availability	The solution is accessible through the ServiceNow cloud instance for authorized users whenever the platform is available.	ServiceNow Cloud Infrastructure
5	Performance	Flow Designer automates approvals, task creation, and notifications, reducing manual coordination and improving request processing efficiency.	Flow Designer and ServiceNow Platform
6	Maintainability	Catalog variables and flow actions can be updated easily when business rules, approval requirements, or fulfillment processes change.	Service Catalog and Flow Designer

Conclusion

The architecture uses ServiceNow as a centralized cloud platform for managing licensed software installation requests. The Service Catalog provides a standardized request interface, while Flow Designer automates approval, task creation, and email notification. ServiceNow tables maintain request and task records, enabling transparent tracking throughout the lifecycle. This architecture reduces manual effort, improves processing speed, and supports compliance with licensing and IT governance requirements.